

## HEC Troubleshooting

This document describes various conditions for the HEC and possible solutions for those conditions.



### NOTICE

**Avoid leaks by checking that the bleeder valves on the front of each pump in the bottom of the HEC are fully closed before powering pumps on for the first time.**

**Table 1:** HEC Troubleshooting

| For this condition:                                                                                                                                     | Follow these steps:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Difficulty navigating or interpreting HEC screens.                                                                                                      | Section 5.3 of <a href="#">Heat Exchanger Theory</a> describes how to navigate the screens. Procedures for setting parameters for the HEC are described in subsequent sections of that document.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| HEC display shows zero cryogen compressor coolant flow despite customer coolant flowing through the HEC and the cryocompressor receiving adequate flow. | Perform a TPS reset. If the cryogen compressor flow continues to show zero, the flowmeter in the HEC may be clogged or defective. It can be visually inspected by swinging the power box out of the way (see applicable steps in Power Box replacement procedure).                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| The HEC display shows dashes instead of values for Gradient or Power Electronics coolant flow.                                                          | <p>Check the red VFD display of the pump on the circuit in question. Make sure the display is not dark and that the circuit breaker above it is not turned off.</p> <p>If the VFD display is turned on, the issue most likely is with the PE or GC pressure sensors:</p> <ul style="list-style-type: none"> <li>• If the HEC display shows either the Gradient or Power Electronics coolant flow value, the sensor for the value that is being shown is functioning. Swap the PE (J4) and GC (J3) pressure sensors at the signal box and see if the values change.</li> <li>• The pressure sensors are FRUs and are listed in the FRU manual.</li> <li>• Check the wire crimps at the connectors of any suspected faulty sensors.</li> </ul> |
| The system displays messages saying that the HEC coolant reservoir is low. The HEC display appears to confirm this.                                     | <p>There are upper and lower pressure-sensitive level sensors mounted on the right side of each reservoir tank. These sensors close (turn on) when the coolant level is high enough over the sensors (usually three inches, but sometimes the height must be as much as five inches).</p> <p>You can measure the switch continuity with a DMM (switch closed, low resistance/switch open, very high resistance).</p> <p>Add coolant to the reservoir until the coolant level is four to five inches above the sensors. See <a href="#">HEC Coolant Fill and Coolant Leak Check</a>. PE or GC should stop reporting the warning when the increase in coolant pressure causes the switches to engage.</p>                                      |

| For this condition:                                                                                              | Follow these steps:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                  | Rarely, the pressure sensors become stuck. Stuck sensors can be freed by tapping them with a finger. Do not use anything with a greater mass to tap the sensors.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| The system displays messages saying coolant resistivity is too low.                                              | <p>The system reports a warning when resistivity falls below 3.5 kOhm-cm and inhibits a scan below 2.5 kOhm-cm.</p> <p>Deionize the coolant to restore the proper resistivity. See <a href="#">Heat Exchange Cabinet Coolant Deionization</a>.</p> <p><b>NOTE:</b> Generally, new systems will register around 5-15 kOhm-cm. However, it is possible for a system to have and display a coolant resistivity between 0.01 kOhm-cm and 1000 kOhm-cm, so it is possible that a new system will have a resistivity lower than 2.5 kOhm-cm and will not be able to scan. the solution is to deionize the coolant.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Performing deionization process, but resistivity value reported from HEC LCD is slowly going down instead of up. | <div>  <div> <b>NOTICE</b><br/> <b>Stop the process! Disconnect the inlet line before the tank overflows.</b> </div> </div> <ul style="list-style-type: none"> <li>The inlet line has been incorrectly connected to the facility coolant discharge and facility coolant is back-flushing into the GC Coolant tank.</li> <li>Review the de-ionization service methods procedure on the service methods DVD and, once the configuration has been corrected, continue with the de-ionization process and confirm during the process that the GC coolant resistivity increases.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| The system displays pressure or RF air temperature warning errors.                                               | <p>Any deviations from specifications for pressure or cooling are stored in the system error log.</p> <p>See the Facility Coolant Requirements section of Direction 5670003 <i>Discovery MR750w 3.0T Preinstallation Manual</i> for the HEC pressure and cooling requirements.</p> <p>If the system returns error 2268904 or 2268905:</p> <ul style="list-style-type: none"> <li>The error is displayed if the temperature exceeds the upper temperature limit or falls below the lower temperature limit. These are <math>\pm 2^{\circ}\text{C}</math> from a set point of <math>18^{\circ}\text{C}</math>.</li> <li>See the HEC block diagram. The temperature sensor that is detecting this error is located in the top manifold of the chilled air HEC blower assembly. This sensor rarely fails.</li> <li>This error occurs commonly if the facility coolant cooling capacity is inadequate. Check the facility parameters on the HEC to make sure they are within specification. Facility coolant that is too cold is almost as problematic as coolant that is too warm. The coolant temperature <b>MUST</b> remain within the required upper and lower window requirements as</li> </ul> |

| For this condition: | Follow these steps:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | <p>stated in the Preinstallation Manual for the system.</p> <p><b>NOTE:</b> See <a href="#">Heat Exchanger Theory</a> for information about navigating the HEC screens.</p> <ol style="list-style-type: none"> <li>1. From the Main Menu of the HEC LED screen, select Monitor → Facility.</li> <li>2. Record the <i>Facility Temperature</i>, <i>Inlet Pressure</i>, and <i>Differential Pressure</i> values.</li> <li>3. See the Facility Coolant Requirements section of Direction 5670003 <i>Discovery MR750w 3.0T Preinstallation Manual</i> for the HEC pressure and cooling requirements.</li> <li>4. If the Facility parameters do not meet specifications, have the customer fix the coolant flow or temperature issue. (The inline filter may require replacement.)</li> <li>5. If the Facility parameters meet specifications, check the mixing valve values             <ol style="list-style-type: none"> <li>i. From the Main screen of the HEC LED display, select Diags → PID Monitor.</li> <li>ii. Record the <i>Blower Mixing Valve Percent Open</i> value.</li> <li>iii. Typically, this valve is at around 50% open. If the value is significantly below 50%, confirm that facility coolant temperature is not too low. If the value is significantly above 50%, check that facility coolant temperature is not too high and also check for adequate blower air flow. It is often necessary to seek assistance from the HVAC team that maintains the chiller to get facility coolant temperature under control. Call the OLC before ordering any parts for this problem.</li> </ol> </li> <li>6. If the facility parameters are within specification and the mixing valve is operating normally:             <ol style="list-style-type: none"> <li>i. Check to see whether the error condition always corrects within 1 to three seconds, or whether a TPS Reset is always needed to correct the error condition.</li> <li>ii. If either of these conditions is true, contact the OLC and request that the <code>cooling.xml</code> file be examined for spiking.</li> </ol> </li> </ol> <p>Refer to <a href="#">HEC Data Viewer</a> for additional information on analyzing trends.</p> <p>Refer to the Lytron vendor manual (GE part number 5161931) for vendor specifications.</p> |

| For this condition:                                                                                                                                                                                                                                                                                                                                                                                                               | Follow these steps:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Suspected bad pressure sensor.                                                                                                                                                                                                                                                                                                                                                                                                    | <p>It is important to determine whether the sensor or the Signal Box (PLC) is actually faulty.</p> <p>If you are testing an HEC facility coolant inlet or outlet sensor:</p> <ol style="list-style-type: none"> <li>1. Note the facility inlet and outlet pressure readings from the HEC LCD screen.</li> <li>2. Swap J5 and J6 on the Signal Box and confirm the facility inlet and outlet pressure readings are swapped on the HEC LCD screen. If they are swapped, the sensor is faulty.</li> </ol> <p>If you are testing a GC pressure sensor:</p> <ol style="list-style-type: none"> <li>1. Note the GC inlet pressure reading from the HEC LCD screen.</li> <li>2. Swap J3 and J6 on the Signal Box and confirm the same pressure reading on the <i>Facility Inlet Pressure</i> on the HEC LCD screen.</li> </ol> <p>If you are trying to test a PE pressure sensor:</p> <ol style="list-style-type: none"> <li>1. Note the PE inlet pressure reading from the HEC LCD screen.</li> <li>2. Swap J4 and J6 and confirm the same pressure reading on <i>Facility Inlet Pressure</i> on the HEC LCD screen.</li> </ol> <p><b>NOTE:</b> Alternatively, you can test GC and PE pressure sensors by swapping J4 with J5 (<i>Facility Outlet Pressure</i>) instead of swapping J4 and J6.</p> |
| <p>One or both of these errors appear repeatedly in error log but scanning not inhibited:</p> <p>Heat Exchanger Cabinet (HEC) detected the gradient coil coolant flow is too low. Verify that the pump is turned on, all inline valves are open.</p> <p>Heat Exchanger Cabinet (HEC) detected the power electronics coolant flow is too low. Verify that the power electronics pump is turned on, all inline valves are open.</p> | <ul style="list-style-type: none"> <li>• Verify that in ready (not standby) mode, HEC VFD PPMP reads 50.</li> <li>• Verify that in ready (not standby) mode, HEC VFD GPMP reads 50. If the value is wrong, check for a bad VFD module.</li> <li>• Verify that PE pressure is 4.2 to 4.8 bar. Verify that GC pressure is 5.2 to 5.3 bar.</li> <li>• If pressure reads 7.6 bar, replace the pressure sensor. Otherwise contact OLC for support.</li> </ul> <p>Do not change any adjustable valve positions for less than maximum flow.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <p>One or both of these errors appear repeatedly in error log but scanning not inhibited:</p> <p>Heat Exchanger Cabinet (HEC) detected the gradient coil coolant flow is too high.</p> <p>Heat Exchanger Cabinet (HEC) detected the power electronics coolant flow is too high. Verify that the power electronics pump is turned on, all inline valves are open.</p>                                                              | <ul style="list-style-type: none"> <li>• Verify that in ready (not standby) mode, HEC VFD PPMP reads 50.</li> <li>• Verify that in ready (not standby) mode, HEC VFD GPMP reads 50. If the value is wrong, check for a bad VFD module.</li> <li>• On the HEC LCD, under Monitor, verify that PE pressure is 4.2 to 4.8 bar. <ul style="list-style-type: none"> <li>◦ If pressure reads 1.0 bar or less, the sensor failed. Replace the sensor.</li> <li>◦ If pressure is greater than 1.0 bar and less than 4.2 bar, the messages can be ignored.</li> </ul> </li> <li>• On the HEC LCD, under Monitor, verify that GC pressure is 5.2 to 5.3 bar.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

| For this condition:               | Follow these steps:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                   | <ul style="list-style-type: none"> <li>◦ If pressure reads 1.0 bar or less, the sensor failed. Replace the sensor.</li> <li>◦ If pressure reads greater than 1.0 bar and less than 5.2 bar, the messages can be ignored.</li> </ul> <p>Do not change any adjustable valve positions for less than maximum flow.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Suspected bad temperature sensor. | <p>It is important to determine whether the sensor or the Signal Box (PLC) is actually faulty.</p> <p>If you are testing the HEC facility coolant inlet sensor:</p> <ol style="list-style-type: none"> <li>1. Note the <i>Facility</i> inlet temperature reading from the HEC LCD screen.</li> <li>2. Note the GC inlet temperature reading from HEC LCD screen.<br/> <b>NOTE:</b> The HEC will stop pumping when the GC sensor is disconnected. Pumps will restart when the sensor is reconnected.</li> <li>3. Swap J18 (Facility) and J17 (GC) on the Signal Box, and confirm that the Facility temperature is now visible for GC on the HEC LCD screen, and that the GC temperature is now visible for <i>Facility</i> on the HEC LCD screen.</li> </ol> <p>If you are testing the GC temperature sensor:</p> <ol style="list-style-type: none"> <li>1. Note the <i>Facility</i> inlet temperature reading from the HEC LCD screen.</li> <li>2. Note the GC inlet temperature reading from the HEC LCD screen.<br/> <b>NOTE:</b> The HEC will stop pumping when the GC sensor is disconnected. Pumps will restart when the sensor is reconnected.</li> <li>3. Swap J18 (Facility) and J17 (GC) on the Signal Box, and confirm that the Facility temperature is now visible for GC on the HEC LCD screen, and that the GC temperature is now visible for <i>Facility</i> on HEC LCD screen.</li> </ol> <p>If you are testing the PE temperature sensor:</p> <ol style="list-style-type: none"> <li>1. Note the <i>Facility</i> inlet temperature reading from the HEC LCD screen.</li> <li>2. Note the PE inlet temperature reading from the HEC LCD screen.<br/> <b>NOTE:</b> The HEC will stop pumping when the PE sensor is disconnected. Pumps will restart when the sensor is reconnected.</li> <li>3. Swap J18 (Facility) and J16 (PE) on the Signal Box, and confirm that the Facility temperature is now visible for PE on the HEC LCD screen, and that the PE temperature is now visible for <i>Facility</i> on HEC LCD screen.</li> </ol> <p>If you are testing the Blower temperature sensor:</p> |

| For this condition:                                                                                                                     | Follow these steps:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                         | <ol style="list-style-type: none"> <li>1. Note the <i>Facility</i> inlet temperature reading from the HEC LCD screen.</li> <li>2. Note the <i>Blower</i> inlet temperature reading from the HEC LCD screen.</li> <li>3. Move the connector at J18 (Facility) to J28 (Blower) on the Signal Box, and confirm that the Blower temperature is now visible for <i>Facility</i> temperature on the HEC LCD screen.</li> </ol> <p><b>NOTE:</b> You can move the connector at J28 to J18 to make the temperature check the other way. However, this may require cutting the tie wraps for the J28 cable to reach the J18 connector.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <p>Circuit breaker trips when power is applied to the HEC. Circuit breaker or fuses in Main Disconnect Panel may trip at same time.</p> | <p>Examine the HEC for short circuits:</p> <ol style="list-style-type: none"> <li>1. LOTO the HEC.</li> <li>2. Open the HEC Power Box, and disconnect pump or blower wiring from the VFD.</li> <li>3. Use a DVM to measure resistance between the pump or blower lines (phase-to-phase) and from each line to ground (phase-to-ground) for short circuits.</li> <li>4. Use a DVM to measure resistance between the VFD input and output connections (phase-to-phase) and from each line to ground (phase-to-ground) for short circuits.</li> <li>5. Use a DVM to measure resistance between the BLW CB circuit breaker input and output connections (phase-to-phase) and each connection to ground (phase-to-ground) for short circuits.</li> </ol> <p>If there are no short circuits observed, order and replace the HEC Power Box.</p>                                                                                                                                                                                                                                                                                                                                                                               |
| <p>There is insufficient water flow to or from components in the PGR cabinet.</p>                                                       | <ol style="list-style-type: none"> <li>1. Check for kinks in the supply and return hoses, especially where hoses are behind units. You can often quickly clear a kink by twisting a pressurized hose a full rotation in one direction and then twisting it in the opposite direction.</li> <li>2. Use a discharge hose and a flask or bucket as troubleshooting tools to check for coolant flow on the supply and return hoses and through components.</li> </ol> <div data-bbox="776 1476 894 1598">  </div> <div data-bbox="899 1476 1146 1524"> <p><b>NOTICE</b></p> </div> <div data-bbox="899 1528 1318 1707"> <p><b>Use the Coolant Draining Kit that is shipped with the HEC (and available as a FRU) to drain PGR modules. Do not use any pump other than the supplied hand pump.</b></p> </div> <div data-bbox="776 1808 894 1929">  </div> <div data-bbox="899 1808 1146 1856"> <p><b>NOTICE</b></p> </div> <div data-bbox="899 1860 1318 1917"> <p><b>The water cooling system for the PGR cabinet is rated to 65 psi.</b></p> </div> |

| For this condition:                                                                                           | Follow these steps:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                               | <p>3. Use a high volume pump to try to clear blockages in components. Try the return side first, then the supply side. You may need to repeat this a few times to get the blockage to clear.</p> <div>  <div> <b>NOTICE</b><br/> <b>Use the Coolant Draining Kit that is shipped with the HEC (and available as a FRU) to drain PGR modules. Do not use any pump other than the supplied hand pump.</b> </div> </div> <p>4. Check that ball valves and check valves in the cooling lines from the HEC are open and installed in the correct orientation. Make sure fittings are fully engaged on each component in the PGR cabinet</p> |
| HEC pumps are constantly cycling on and off.                                                                  | HEC does this to protect pumps from overheating and damage due to operation while the pump is dry. Check for any coolant line obstructions (kinks, twists, etc.) or improper installations (check valve installed backwards) or closed ball valves (tank valve, external valves, etc.).                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Need to view HEC parameters in real-time.                                                                     | <p>The system uses event-based polling to monitor the HEC, storing the data it receives to the <code>cooling.xml</code> log file.</p> <p>Open a c-shell and type:</p> <pre>more /usr/g/service/log/paramdata/cooling.xml</pre> <p>Each output describes units in which the value is expressed. You must pay attention to the units and scaling.</p> <p>Refer to <a href="#">HEC Data Viewer</a> for additional information on analyzing trends.</p>                                                                                                                                                                                                                                                                     |
| At power up, the GC/PE coolant pump or blower is not operating. The VFD for the component displays --- or UF. | <p>It is possible that the Run command is not being sent from the signal box.</p> <ol style="list-style-type: none"> <li>1. On the signal box controller, hold the <b>Up</b> arrow while you press &lt;F1&gt; to toggle the blower on, &lt;F2&gt; to toggle the GC pump on, and &lt;F3&gt; to toggle the PE pump on.</li> <li>2. If the VFDs do not ramp up to the proper speed, check the connections at the side of the signal box, and check for a bad VFD.</li> <li>3. If all appears okay, replace the signal box.</li> </ol>                                                                                                                                                                                      |
| After initial power up, the system reports the GC coolant resistivity is out of range (too low or too high).  | <p>If the GC resistivity value is less than 2.5 kOhm-cm and greater than 1 kOhm-cm, deionize the coolant. See <a href="#">Heat Exchange Cabinet Coolant Deionization</a>.</p> <p>A GC resistivity value greater than 20.0 kOhm-cm may mean an FE forgot to add rust inhibitor and biocide chemicals after performing the last de-ionization procedure. Perform de-ionization again and</p>                                                                                                                                                                                                                                                                                                                              |

| For this condition: | Follow these steps:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | <p>insure these are added per procedure. If values are unchanged after adding additives then proceed to "If Fuse OK" section below.</p> <p>A GC resistivity value less than 1 kOhm-cm may indicate faulty hardware:</p> <ul style="list-style-type: none"> <li>• Check the PLC code version: <ol style="list-style-type: none"> <li>1. From the main Signal Box PLC screen, press <i>Diags</i>. The code version is a date code and will look like<br/>G3.MM.DD.YYYY.</li> <li>2. If the code version is anything earlier than G3.02.16.2009 and the HEC only has one DI sensor in the GC tank, wait until the MR host computer has been connected to the HEC and the first successful TPS Reset is completed. This will download version G3.02.16.2009 or newer.</li> <li>3. Recheck resistivity value.</li> </ol> </li> <li>• Check the condition of the DI circuit board fuse. <ol style="list-style-type: none"> <li>1. Remove the HEC Signal Box cover and check the condition of the fuse located in the upper left corner of the DI circuit board. <p><b>NOTE:</b> A spare fuse is usually provided in a plastic bag inside the Signal Box. If the fuse is unavailable, order 5266827 Fuse FRU Kit.</p> </li> <li>2. If the fuse is open: <ol style="list-style-type: none"> <li>i. View the voltage selector toggle switch setting through the view holes on front of the HEC Power Box (this may require a flashlight).</li> <li>ii. Confirm that the switch is in the correct position for incoming voltage to the HEC. See Section 4.3 of <a href="#">Power-On Sequencing</a>.</li> </ol> </li> <li>3. If the fuse is intact: <ol style="list-style-type: none"> <li>i. Note the resistivity value reported for the Gradient circuit cooling circuit on the HEC LED screen.</li> <li>ii. Disconnect the resistivity sensor DI1 from J23 on the side of the Signal Box.</li> <li>iii. Recheck the resistivity value on the HEC LED screen. It should read 0.01 kOhm-cm.</li> <li>iv. If removing resistivity sensor DI1 does not result in a significant change in the resistivity value, replace the Signal Box. <p><b>NOTE:</b> During operation, a good working DI sensor will measure approximately 1.2 VDC (corresponds to ~12.0 kOhm-cm resistivity) across the white (+) and black wires</p> </li> </ol> </li> </ol> </li> </ul> |



| For this condition:                                                        | Follow these steps:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                            | <p>(-). A faulty, open sensor will measure ~2.5 VDC. A faulty, shorted sensor will measure much less than 1.2 VDC.</p> <p>Early units have a second sensor (J10 on the Signal Box), later units have only a single sensor. The second sensor is not measured (not used) in systems with 20.0 M4 software and later (Signal Box PLC code versions G3.02.16.2009 and newer. Check the code revision on HEC Signal Box PLC by pressing <i>Dia</i>gs on the Main Screen).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| The cable dressing for the HEC Signal Box appears different.               | <p>There are two versions of the cable dressing for the HEC Signal Box. The newer version is used in HEC version 11 and higher and Signal Box version 2 and higher.</p> <p>To determine the version of the HEC, locate the rating label for the HEC behind the left side front door on the floor of the cabinet.</p> <p>To determine the version of the Signal Box, locate the label on the side of the Signal Box. If there is no label, the Signal Box is an early model.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| The mixing valve reading on the Signal Box LCD screen is at 100% or at 0%. | <p>A reading of 100% means:</p> <ul style="list-style-type: none"> <li>• The maximum amount of coolant is being passed through the heat exchanger.</li> <li>• The stem on the mixing valve is completely down so that the stem is almost flush with the mixing valve case.</li> <li>• The PLC PID value for the mixing valve position is 4094 (this value is not displayed).</li> </ul> <p>A reading of 0% means:</p> <ul style="list-style-type: none"> <li>• The minimum amount of coolant is being passed through the heat exchanger (the maximum amount of coolant is being passed around the heat exchanger).</li> <li>• The stem of the mixing valve is at the maximum height above the mixing valve case.</li> <li>• The PLC PID value for the mixing valve position is 0 (this value is not displayed).</li> </ul> <p>Mixing valve positions (0% to 100%) does not have any significant effect on total facility coolant flow through HEC, because these are three-way valves so the same amount of coolant passes through the cabinet regardless of valve position.</p> |
| HEC Blower circuit breaker trips and needs to be manually reset.           | The blower may be drawing excessive current because of a blockage.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| For this condition:                                                                                               | Follow these steps:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                   | <ul style="list-style-type: none"> <li>• Check the condition of the air filter on the PEN wall in the scan room. Clean or replace if it is dirty.</li> <li>• Inspect the area where the filter is located and make certain that it is not too restricted so that air flow into the filter is poor.</li> <li>• Inspect blower motor or hosing inside the HEC for evidence of excessive amounts of dust or lint.</li> </ul> <p>Additionally, there are hoses internal to the HEC that connect to the blower. It may be necessary to remove the suction hose (the one not connected to the liquid to air heat exchanger) and check if the internal wall of the hose is still intact and not delaminated or clogging airflow.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Coolant to PE or GC components is too cold or warm (HEC does not appear to be regulating temperature of coolant). | <ul style="list-style-type: none"> <li>• Inspect the top of the mixing valves and make certain that nothing like hose, cable, etc. is depressing white plastic manual override button on top of mixing valve.</li> <li>• Check mixing valve percentages from HEC LCD screen. Contact OLC if necessary for support.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Need to manually adjust a mixing valve                                                                            | <ol style="list-style-type: none"> <li>1. Obtain a 4mm or 5/32" hex Allen wrench.</li> <li>2. While holding the manual override button (labeled <i>PRESS</i>) flush with the case, insert the Allen wrench into the recessed manual override actuator adjust screw and rotate (small rotational force required) until the desired valve position is reached.</li> <li>3. The position indicator hex nut will move vertically up or down in response to this adjustment. It does not rotate.</li> </ol> <div data-bbox="776 1163 894 1283"> </div> <div data-bbox="899 1163 1146 1209"> <p><b>NOTICE</b></p> </div> <p><b>Do not attempt to adjust or rotate the position indicator hex nut. It is a visual indicator only.</b></p> <ul style="list-style-type: none"> <li>◦ If the position indicator nut is fully up, the valve is at 0% heat exchanger coolant flow (minimum cooling).</li> <li>◦ If the position indicator nut is fully down (bottom part of nut flush with case), the valve is at 100% heat exchanger coolant flow (maximum cooling).</li> </ul> <ol style="list-style-type: none"> <li>4. Select [ESC] on the keypad until the HEC LCD is at the main menu.</li> <li>5. Select <b>DIAGS, PID</b> to reach the screen labeled <i>PID MONITOR</i>.</li> <li>6. On the PID MONITOR screen: <ul style="list-style-type: none"> <li>◦ The left-most temperature value is actual measured coolant (GR COIL or PR ELEC) or air temp (BLOWER).The left-most temperature value will not start to change until about 1 minute after making significant mixing valve position change.</li> <li>◦ The center temperature is set-point, target-value.</li> </ul> </li> </ol> |

| For this condition:                                         | Follow these steps:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                             | <ul style="list-style-type: none"> <li>◦ The right-most percentage value is amount of facility coolant PLC is commanding through heat exchanger. After the manual override button is released, the Signal Box PLC will begin slowly repositioning the mixing valve to match the measured temperature with the target value. The vertical position of the position indicator hex nut changes at a rate of 0.1mm per second. Be patient and watch carefully. The mixing valve percentage demand does not change in response to manual adjustment. This is demand requested by the Signal Box PLC, not actual measured value</li> </ul>                                                                                                                                                                                                                                                                                                                                 |
| On power up, VFD displays two-letter code such as <b>UF</b> | <ol style="list-style-type: none"> <li>1. On the signal box controller, hold the <b>Up</b> arrow while you press &lt;F1&gt; to toggle the blower on, &lt;F2&gt; to toggle the GC pump on, and &lt;F3&gt; to toggle the PE pump on.</li> <li>2. If there is no response, cycle HEC power and try again to start the pumps manually (<a href="#">Step On the signal box controller, hold the Up arrow while you press F1 to toggle the blower on, F2 to toggle the GC pump on, and F3 to toggle the PE pump on.</a>).</li> <li>3. If you cannot clear the code, try these solutions: <ul style="list-style-type: none"> <li>◦ Check for proper incoming AC mains power.</li> <li>◦ If the code is visible only on a single VFD, swap its control wiring with that of neighboring unit and see if the problem moves.</li> <li>◦ If the same code is visible on all VFDs, and incoming AC mains power is okay, check operation of the Signal Box.</li> </ul> </li> </ol> |
|                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

▲  
Back to top